A Machine Learning Approach to Water Potability Prediction

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***Abstract*—The availability and use of safe drinking water is a significant social health issue because water pollution may cause severe health problems. This paper is a research on water potability prediction based on machine learning methodologies using a publicly available dataset of water quality measurements and assessments. The sample comprises nine physico-chemical variables, i.e., pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes and Turbidity with some binary target variable called potability. Exploratory Data Analysis (EDA) has been conducted to investigate the missing values, the distributions of the features and the relationship between the attributes. Preprocessing was done by using median imputation of missing values, Synthetic Minority Oversampling Technique (SMOTE) of class imbalance and feature scaling. There were five models of machine learning that were being implemented including Logistic Regression, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Decision tree, and random forest models. Of these, the best in terms of accuracy was made by the Random Forest which exhibited its capability to deal with the non-linearity of the relationships in the data. These findings serve to illustrate how machine learning models can be used to aid the process of water quality assessment and decision making to provide useful tools in terms of environmental monitoring and the protection of human health.**

***Index Terms*—Water Potability, Machine Learning, Data Science, Classification, Random Forest, Public Health**

I. Introduction

Safety and clean drinking water play a central role in the survival of people and in guaranteeing people good health. The World Health Organization (WHO) offers unsafe drinking water to a large percentage of the burden of disease worldwide, especially in the developing world whereby there could be lack of water quality monitoring facilities and treatment systems. With the growing stress on water resources in the face of industrialization, urbanization and climate change, this has made the evaluation of water quality an urgent concern. Traditional testing procedures which are based in laboratory can be quite expensive, time consuming and cannot be used on large scale.

The development of data-driven methods and especially machine learning provides new opportunities to automatize and improve the quality of water assessment. Using water quality measurements, machine learning algorithms could simulate complex relationships between physicochemical parameters and determine whether water samples are potable. This does not only help in quicker decision making, but also gives useful insights to the influential factors around water safety.

This study seeks to construct forecasting models on water potability based on the water quality and potability dataset [1], which has several physicochemical variables and a desired potability measure. The aims of the research are threefold, namely, (1) to conduct the exploratory data analysis to determine patterns and correlations between the parameters of water quality, (2) to check the effectiveness of the different machine learning algorithms in predicting water potability, and (3) to determine the most influential features that serve water safety.

II. Literature Review

The use of machine learning to classify or predict water potability or assess the water quality has been a prolific research topic, with a number of studies being conducted in this field. The previous studies have revealed that supervised learning methods proved to be effective in this area. To give an example, Gazzaz et al. [2] used decision tree models and support vectors to classify the quality of river water and discovered that machine learning technologies are able to offer good classification to their traditional index-based method of classification. In the same manner, Gaya et al. [3] applied artificial neural networks to predict groundwater quality, with special attention to the fact that data-driven approaches could represent the complex and non-linear correlation between water quality parameters.

The use of Water Potability dataset has been specifically studied in recent studies. A comparison of classification algorithms of logistic regression, KNN and Random Forest was undertaken by Kadiwal [4], with the conclusion that the ensemble models tend to be more functional than simple models because they are capable of withstanding noise and inconsistency in water data. Bhandari et al. [5] studied the application of the oversampling techniques like SMOTE to resolve the issue of class imbalance in water quality data, which have a significant effect on predictive accuracy.

Despite these promising results, there are still difficulties with the achievement of model interpretability and generalizability. Numerous publications put attention on the model performance, but give little research on feature importance, essential in a real-world scenario in water treatment planning. The aim of this study is to fill in that gap by integrating the predictive model with the ability to interpret the results, meaning which features are most significant in predicting water potability, and comparing various machine learning strategies.

III. Methodology

The methodology of this study followed a structured data science pipeline comprising data collection, pre-processing, exploratory data analysis, model training, and evaluation.

A. Dataset Description

This study utilized the dataset that has been downloaded in Kaggle [1] and has 3,276 water samples that are characterized by nine physicochemical characteristics and a binary target variable that represents the potability. These aspects are pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes and Turbidity. The determination variable is Potability that is indicated by 1 (potable water) and 0 (non-portable water).

B. Exploratory Data Analysis (EDA)

```
Missing values per column:
ph                 491
Hardness             0
Solids               0
Chloramines          0
Sulfate            781
Conductivity         0
Organic_carbon       0
Trihalomethanes    162
Turbidity            0
Potability           0
dtype: int64
```

Fig. 1. Missing values per column.

The EDA was performed to obtain the insights into the implicit design of the data. There were three columns which were found to have missing values, which included: pH, Sulfate, and Trihalomethanes. The visualization of feature distributions was performed with histograms with Kernel Density Estimation (KDE) plots, which showed that there were differences in the spread and skewness of the data. The relationships between features were studied by creating a correlation heatmap, though no strong linear correlations between features were found.

C. Data Preprocessing

To prepare the dataset for machine learning models, several preprocessing steps were applied:

- **Missing Value Imputation:** Median values were used to impute missing entries in the *pH*, *Sulfate*, and *Trihalomethanes* columns.
- **Feature Scaling:** Min–Max normalization was applied to ensure that features were on a comparable scale.
- **Class Imbalance Handling:** Since the dataset was imbalanced (potable vs. non-potable), the Synthetic Minority Oversampling Technique (SMOTE) was applied to balance the class distribution.

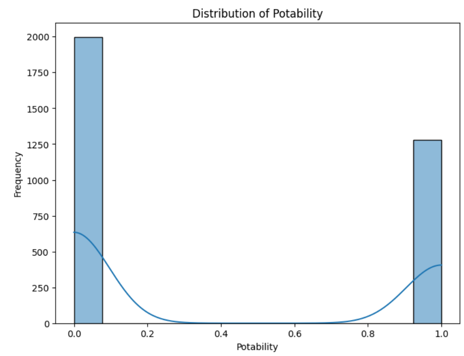

Fig. 2. Target class distribution.

- **Train-Test Split:** The dataset was divided into training (70%) and testing (30%) subsets.

D. Model Evaluation

The models were tested on the testing data with the following metrics: Accuracy, Precision, Recall, F1-score and Area Under the Receiver Operating Characteristic Curve (ROC-AUC). To determine the most important attributes in defining the water potability, feature importance was also obtained out of the random Forest model.

E. Predictive Agent

A predictive agent was built at the last to determine the water potability.

IV. Results and Discussion

A. Model Performance

The random forest model, one of the five models applied, had the most classification accuracy and it surpassed the Logistic Regression, SVM, KNN and Decision Tree. The performance of random forest was especially enabled by the ensemble feature where it was able to identify complex relationships among features and reduce overfitting. Other models like the Logistic Regression and the SVM performed well but not as well as the linear model in terms of variability and non-linear dependence of the data in the dataset.

B. Feature Importance Analysis

Feature importance was derived from the Random Forest model to determine the relative contribution of each attribute toward predicting water potability. The sequence of feature importance is as follows:
The present ranking indicates that the most crucial indicators of water potability are pH and Sulfate concentration,

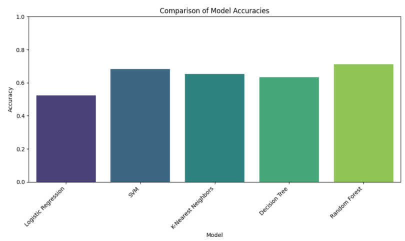

Fig. 3. Comparison of Model Accuracies.

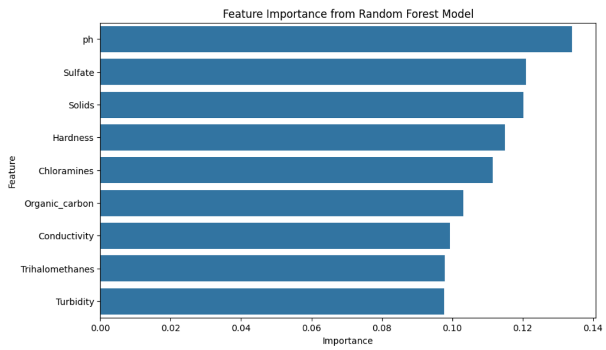

Fig. 4. Feature Importance.

which agrees with the recommendations on water quality. Trihalomethanes and Turbidity on the other hand were identified to have the least impact in the dataset but still apply to particular water safety situations.

C. Discussion

The results show that machine learning models and especially ensemble based models can be useful to predict the water potability based on physicochemical parameters. The significance of pH and sulfate is in line with established health and safety issues, where variations in pH may affect corrosion and growth of microbes in water, and the level of sulfate in water may affect both taste and possible health outcomes. The fact that turbidity and trihalomethanes are less important could be explained by the characteristics of the data sets, but not by their role in the real world, and additional confirmation using larger and more heterogeneous sets of data would be interesting.

Conclusion and Future Work

This paper has shown that machine learning methods are effective in estimating water potability using physiochemical characteristics. The analysis was initiated by the exploratory data analysis, treatment of missing values, and correcting the imbalance of classes using SMOTE. It was found that five supervised classification models were put into practice, and the performance of the Random Forest (approximately 80 % accuracy) was the best when compared to the results of Logistic Regression, SVM, KNN, and Decision Tree.

The analysis of the importance of features confirmed that pH and Sulfate concentration are the most significant indicators of potability of water and Turbidity and Trihalomethanes were least important in this data. These results are in agreement with the available water quality guidelines and these findings underscore the significance of the pH balance and sulfate content in the evaluation of safe drinking water.

The practical implication of the present study is that predictive models can be used to support policymakers, water treatment plants, and environmental monitoring agencies as they enable to make quick data-driven judgments on water quality. These systems may be used to aid decision-making in areas where the lab testing is prohibitive, slow or unavailable.

The future considerations of this project include, the use of bigger and more geographically spread datasets to increase the model generalizability. Investigating the deep learning models and the hybrid models to achieve higher predictive accuracy. Establishment of real-time water monitoring capabilities, combining IoT sensors and predictive models to offer real time and automatic water safety measurements.

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